Sasini Wanigathunga

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Education

University of Moratuwa, Sri Lanka 2021 - 2025

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering ***GPA: 3.90/4.0***

- **Dean's List:** Semesters 1, 2, 3, 4, 6, 7 and 8
- **Selected Coursework:** Deep Learning for Vision, Image Processing and Machine Vision, Pattern Recognition, Introduction to Engineering Optimization, Data Structures and Algorithms, Applied Statistics, Linear Algebra, Calculus, Neural Networks and Fuzzy Logic, Security in Cyber-Physical Systems

Sujatha Vidyalaya, Matara, Sri Lanka 2011 - 2019

GCE Advanced Level - Physical Science Stream | District Rank: 7 | Island Rank: 97 | 3As ***Z-score - 2.6629***

- Combined Mathematics, Physics, Chemistry

Relevant Coursework

Coursera

Mastering REST APIs with FastAPI (Packt, Coursera)
Machine Learning in the Enterprise (Machine Learning on Google Cloud Specialization - Google Cloud)
Retrieval Augmented Generation (RAG) (DeepLearning.AI)
Machine Learning Specialization (DeepLearning.AI, Stanford University)
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Experience

AI Research Engineer | Robotic Assistance Devices, USA Mar 2025 - Present

- Developed browser-based automation for security operations, including dashboard navigation and interaction workflows.
- Developed automated camera-health monitoring and focus correction using image blur scores.
- Developed agentic workflows for a voice assistant and a multi-agent vision-language model (VLM) harness.
- Deployed and evaluated machine learning models on AWS using SageMaker, EC2, and S3, assessing inference performance and scalability.
- Developed a retrieval-augmented generation (RAG) system for operational log analysis and natural-language security incident investigation.
- Conducted research and development on video-based fall detection and keyframe extraction.

Machine Learning Engineer - Intern | FcodeLabs, Sri Lanka Nov 2023 - May 2024

- Worked on privacy-focused ML solutions, including developing a novel method for removing PII from speech data and building production-ready NLP modules for privacy data masking and LLM-based activity suggestion features.

Visiting Instructor | University of Moratuwa, Sri Lanka Feb - Apr 2024

- Worked as a visiting instructor for EN1094: Laboratory Practice

Publications

Test-Time Optimization for Domain Adaptive Open Vocabulary Segmentation Neural Networks, 2027

U De Silva*, D Samaraweera*, S Wanigathunga*, K Kariyawasam*, K Ranasinghe, M Naseer, R Rodrigo

* Equal contribution

Skills

Programming Languages & Tools: Python, MATLAB, C++, C#, Git, Playwright, Unity, FFmpeg
AI/ML: PyTorch, TensorFlow, Hugging Face, NeMo, OpenCV, spaCy, NLTK, LangChain, LangGraph, vLLM
Deployment & Cloud: ONNX Runtime, TensorRT, Docker, AWS (SageMaker, EC2, S3, CloudWatch)
Sports: Chess (*FIDE Rating - 1531*)
Languages: English (Professional proficiency), Sinhala (Native proficiency)

Awards

Prof. C. Patuwathavithane Memorial Award | Academic Year 2023/2024 *Sep 2026*

Prof. K.K.Y.W. Perera Award | Academic Year 2023/2024 *Sep 2026*

World Rank 6 in IEEE SPS Video and Image Processing (VIP) Cup 2023 *Aug - Sep 2023*

IEEEXtreme 17.0 | World Rank: 389 | Country Rank: 26 *Oct 2023*

IEEEXtreme 16.0 | World Rank: 874 | Country Rank: 61 *Oct 2022*

Selected Projects

Vision Language Models - Final Year Project | *Computer Vision, VLMs, Prompt Learning* Jun 2024 - Present
- Developed a novel test-time optimization framework. Evaluated across 22 challenging domain specific datasets and obtained overall +2.03 mIoU improvement over the state-of-the-art.

Spoken Named Entity Localization | *NLP, NER, Spoken Language Understanding* Mar 2024 - Present
- Developed a novel approach for text-independent PII removal in speech data. Achieved +4.1 frame-level F1 score, while using 67% fewer parameters and low latency over the state-of-the-art.

ProjectPulseAI | *LLMs, Groq, RAG, Next.js, Jira API, AstraDB* Jan 2025
- Developed a LLM-powered application for project managers using Retrieval-Augmented Generation (RAG) to integrate project documents, company resources, web links, and Jira issue tracking.

Ophthalmic Biomarker Detection | *Computer Vision, Deep Learning* Aug - Oct 2023
- Optimized the algorithm to predict the presence or absence of biomarkers on OCT scan images. Tested with different preprocessing techniques and backbones and optimized the model by changing the classification layers.

Software Design Competition | *Unity, C# Programming* Jan - July 2024
- Designed a game using Unity with WebGL including API authentication, player profile, a questionnaire that can be opened using a web browser, dynamic and interactive game environment and a leaderboard.

Leadership Experience

IEEE Industrial Electronics Society Student Branch Chapter - University of Moratuwa
- Secretary, Chief Editor, Assistant Editor, Member of Public Relations Team

Electronic Club - University of Moratuwa
- Project Chairperson, Student Editor-in-Chief, Editor, Sub Editor

AIESEC in Colombo South, Sri Lanka
- Content Specialist in Public Relations and Marketing Team, iGT International Relations and Matching Team

References

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